

Graphing Quadratics Test

Algebra 2 - On-Level

Instructions: Show all work for full credit. Write answers clearly in the spaces provided.

Total Points: 100 Time: 45 minutes

Formula Reference:

- Vertex Form: $y = a(x - h)^2 + k$ where vertex is (h, k)
- To complete the square: Add and subtract $\left(\frac{b}{2}\right)^2$

Part 1: Identifying Key Features from Vertex Form (25 points)

1. **(5 points)** For the equation $y = (x - 3)^2 + 5$:

- a) Vertex: _____
- b) Axis of symmetry: _____
- c) Does it open up or down? _____

2. **(5 points)** For the equation $y = -2(x + 4)^2 - 1$:

- a) Vertex: _____
- b) Axis of symmetry: _____
- c) Does it open up or down? _____

3. **(5 points)** For the equation $y = \frac{1}{2}(x - 1)^2 + 3$:

- a) Vertex: _____
- b) Is the parabola wider or narrower than $y = x^2$? _____
- c) Does it open up or down? _____

4. **(5 points)** Match each equation to its description. Write the letter in the blank.

_____ $y = (x + 2)^2 - 3$	A. Opens down, vertex at $(1, 4)$
_____ $y = -(x - 1)^2 + 4$	B. Opens up, vertex at $(-2, -3)$
_____ $y = 3(x - 5)^2$	C. Opens up, narrower, vertex at $(5, 0)$

5. **(5 points)** Circle the correct answer for each statement:

- a) The vertex of $y = (x - 7)^2 + 2$ is at: $(7, 2)$ $(-7, 2)$ $(7, -2)$
- b) The parabola $y = -4(x + 1)^2 - 3$ opens: UP DOWN
- c) The axis of symmetry for $y = (x + 5)^2 - 6$ is: $x = 5$ $x = -5$ $x = -6$

Part 2: Converting to Vertex Form (30 points)**Hint:** To complete the square for $y = x^2 + bx + c$:

1. Take half of b : $\frac{b}{2}$
2. Square it: $\left(\frac{b}{2}\right)^2$
3. Add and subtract this value
4. **(10 points)** Convert $y = x^2 + 6x + 5$ to vertex form. *Follow the steps below:*

Step 1: What is half of 6? _____ What is it squared? _____

Step 2: Add and subtract this inside: $y = (x^2 + 6x + \underline{\hspace{1cm}}) - \underline{\hspace{1cm}} + 5$ Step 3: Factor the perfect square: $y = (x + \underline{\hspace{1cm}})^2 + \underline{\hspace{1cm}}$ **Vertex form:** _____**Vertex:** _____

5. **(10 points)** Convert $y = x^2 - 4x + 1$ to vertex form. Show your work.

Vertex form: _____**Vertex:** _____

6. **(10 points)** Convert $y = x^2 + 10x + 20$ to vertex form. Show your work.

Vertex form: _____

Vertex: _____

Part 3: Describing Transformations (20 points)**Reference:** For $y = a(x - h)^2 + k$:

- h shifts the graph left/right ($h > 0$ means right, $h < 0$ means left)
- k shifts the graph up/down ($k > 0$ means up, $k < 0$ means down)
- $a < 0$ reflects over the x-axis (opens down)
- $|a| > 1$ makes it narrower; $0 < |a| < 1$ makes it wider

7. **(7 points)** Describe how $y = (x - 4)^2 + 2$ is transformed from $y = x^2$.

- Horizontal shift: _____
- Vertical shift: _____
- Opens up or down: _____

8. **(7 points)** Describe how $y = -\frac{1}{3}(x + 2)^2 - 1$ is transformed from $y = x^2$.

- Horizontal shift: _____
- Vertical shift: _____
- Reflection (yes/no): _____
- Wider or narrower: _____

9. **(6 points)** Complete the table comparing two parabolas:

Feature	$y = (x - 1)^2 + 3$	$y = (x - 1)^2 - 2$
Vertex		
Axis of symmetry		
Opens up or down		

What do these two parabolas have in common? _____

Part 4: Writing Equations and Application (25 points)

10. (5 points) Write the equation of a parabola in vertex form with:

- Vertex at $(2, -5)$
- Opens upward
- Same width as $y = x^2$

Equation: _____

11. (5 points) Write the equation of a parabola in vertex form with:

- Vertex at $(-3, 4)$
- Opens downward
- Same width as $y = x^2$

Equation: _____

12. (15 points) **Application Problem:** A ball is thrown upward. Its height h (in feet) after t seconds is given by:

$$h(t) = -16t^2 + 32t$$

a) Complete the square to convert this to vertex form. (Show your work)

Hint: Factor out -16 first: $h(t) = -16(t^2 - 2t)$

Vertex form: _____

b) What is the vertex? _____

c) What is the maximum height of the ball? _____

d) At what time does the ball reach its maximum height? _____

Bonus Question (5 points extra credit)

13. **(5 points)** A parabola has vertex at $(-1, 3)$ and passes through the point $(0, 5)$.

a) Write the general form: $y = a(x - (\underline{\hspace{1cm}}))^2 + \underline{\hspace{1cm}}$

b) Substitute the point $(0, 5)$ to find a : $5 = a(0 - (\underline{\hspace{1cm}}))^2 + \underline{\hspace{1cm}}$

c) Solve for a : _____

d) Write the complete equation: _____

END OF TEST

You've got this! Show your work and do your best!